DIABETES CANADA

2018 Clinical Practice Guidelines

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Hyperglycemic Emergencies in Adults

Chapter 15

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2018

Key Changes

- New information on
  - **Diabetic ketoacidosis with SGLT2 inhibitor therapy**

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Hyperglycemic Emergencies

- DKA = **Diabetic Ketoacidosis**
- HHS = **Hyperosmolar Hyperglycemic State**
- **Common features:**
  - Insulin deficiency → hyperglycemia → urinary loss of water and electrolytes

    → **Volume depletion + electrolyte deficiency + hyperosmolarity**
  - Insulin deficiency (absolute) + increased glucagon

    → **Ketoacidosis** (in DKA)

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Suspect DKA or HHS in an ILL Patient with Hyperglycemia (usually)

DKA

- Ketoacidosis
- ECFV contraction
- Milder hyperosmolarity
- Normal to high glucose
- May have ↓LOC
- Beware hypokalemia
- Must use insulin
- Absolute insulin deficiency + increased glucagon

HHS

- Minimal acid-base problem
- ECFV contraction
- Hyperosmolarity
- Marked hyperglycemia
- Marked ↓↓LOC
- Beware hypokalemia
- May need insulin
- Relative insulin deficiency

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DKA, diabetic ketoacidosis; *ECFV*, extracellular fluid volume; *HHS*, hyperosmolar hyperglycemic state; *LOC*, level of consciousness

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Suspect DKA if……

- pH ≤7.3
- Bicarbonate ≤15 mmol/L
- Anion gap >12 mmol/L

  = Serum sodium – (chloride + bicarbonate)
- Positive serum or urine ketones
- Plasma glucose ≥14 mmol/L (but may be lower)
- Precipitating factor

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DKA, diabetic ketoacidosis

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Clinical presentation of DKA

	Symptoms	**Signs**
Hyperglycemia	polyuria, polydipsia, weakness	ECFV contraction
Acidosis	air hunger, nausea, vomiting and abdominal pain altered sensorium	Kussmaul respiration, acetone-odoured breath altered sensorium
Precipitating condition	See list of conditions Slide 20	

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DKA, diabetic ketoacidosis

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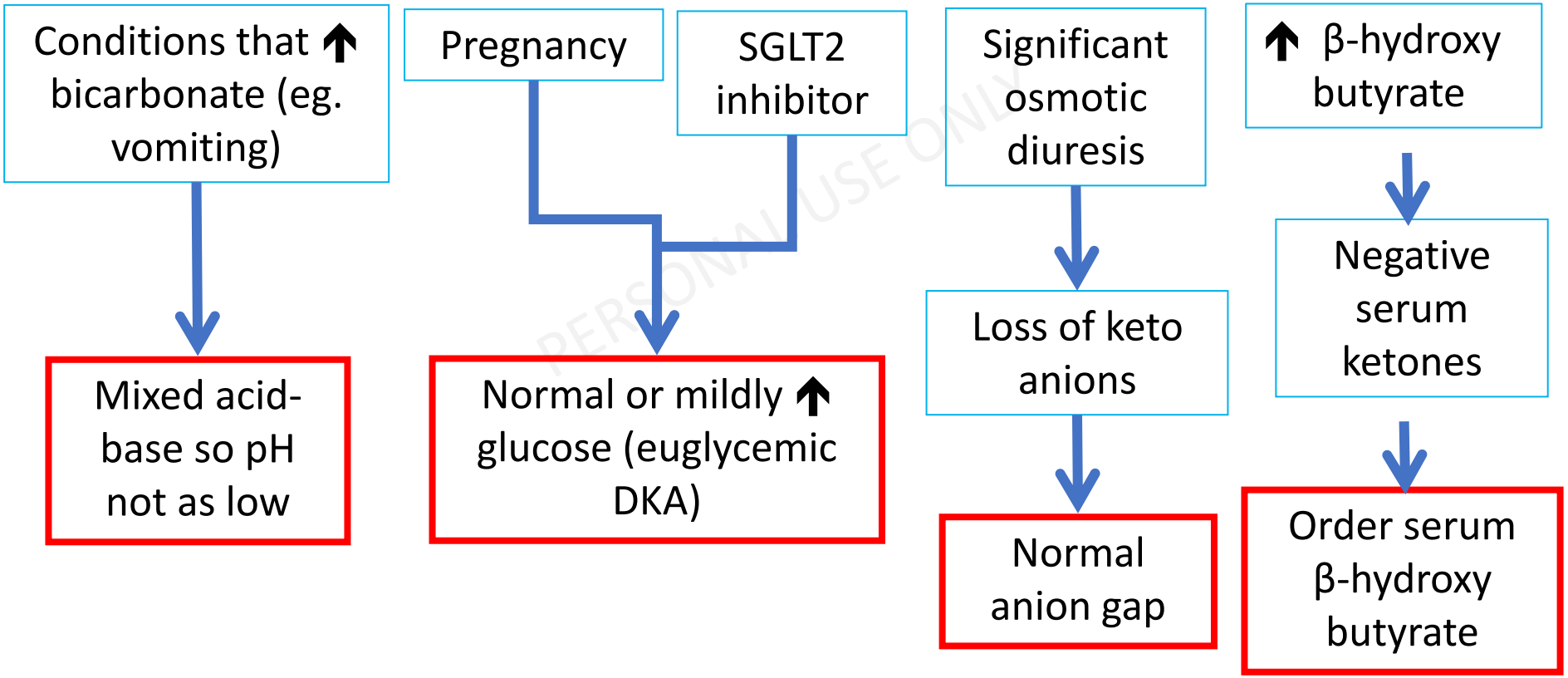
Be Aware of Conditions that may make DKA Diagnosis Difficult



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DKA, diabetic ketoacidosis

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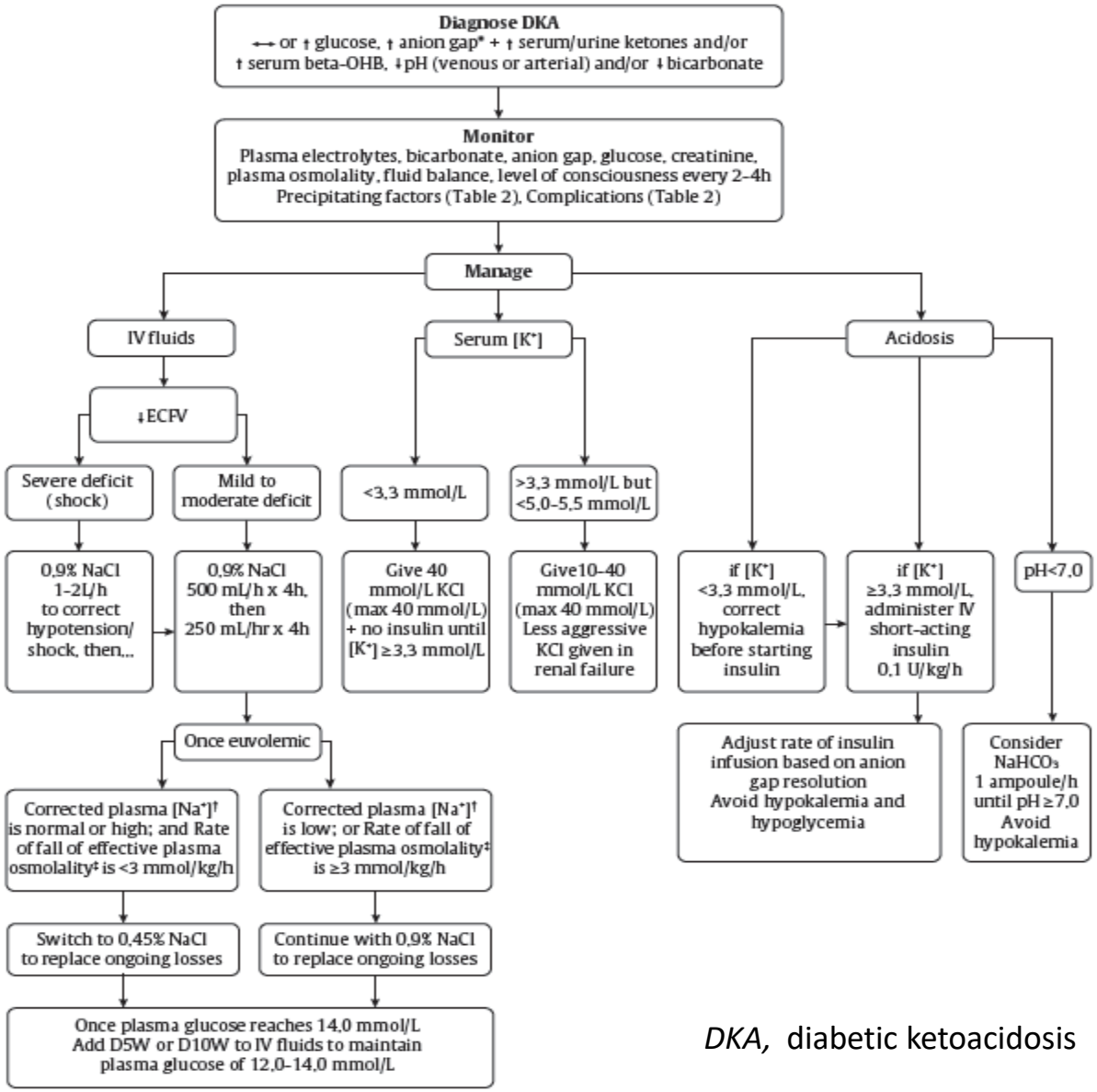
Management of DKA in Adults



DKA, diabetic ketoacidosis

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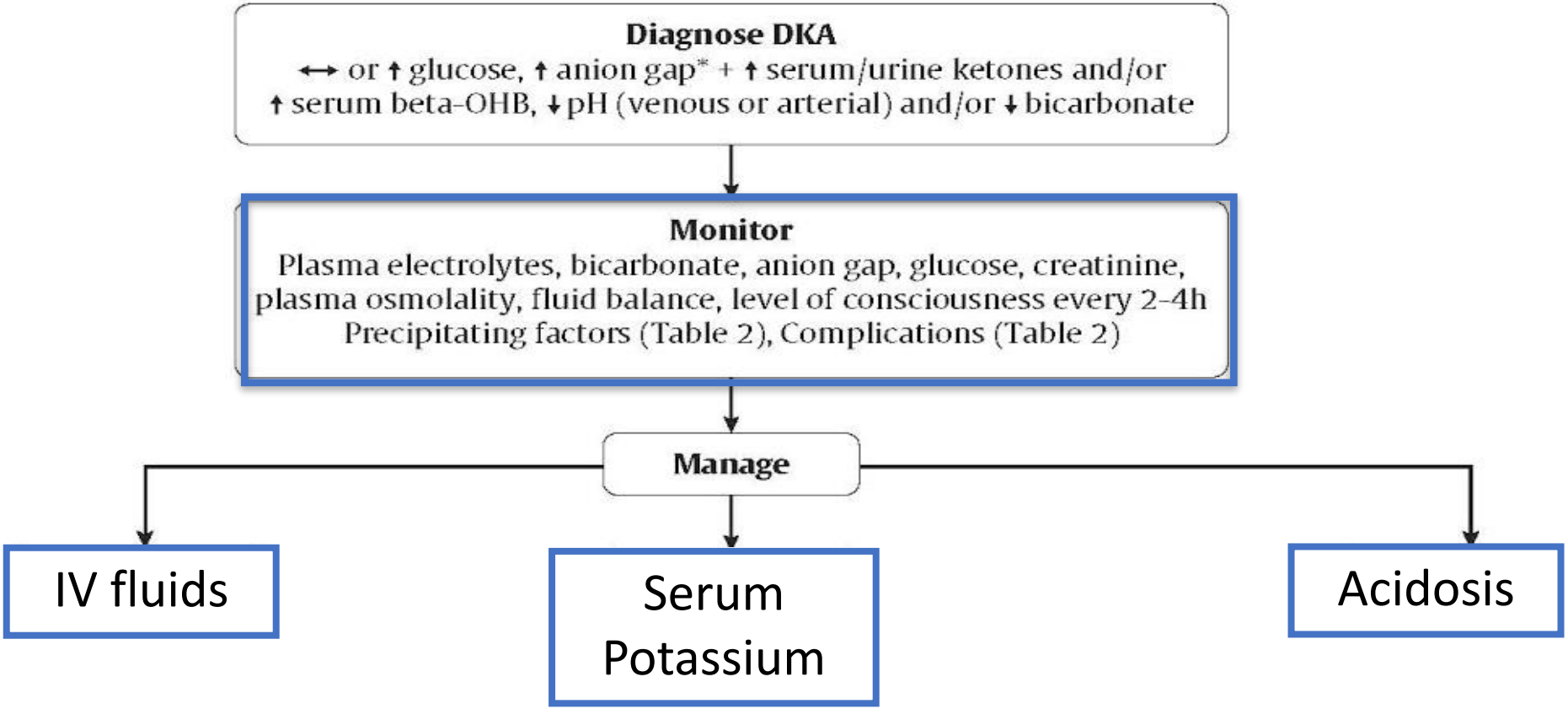
Fluids, Potassium, Acidosis are the Pillars of Treatment



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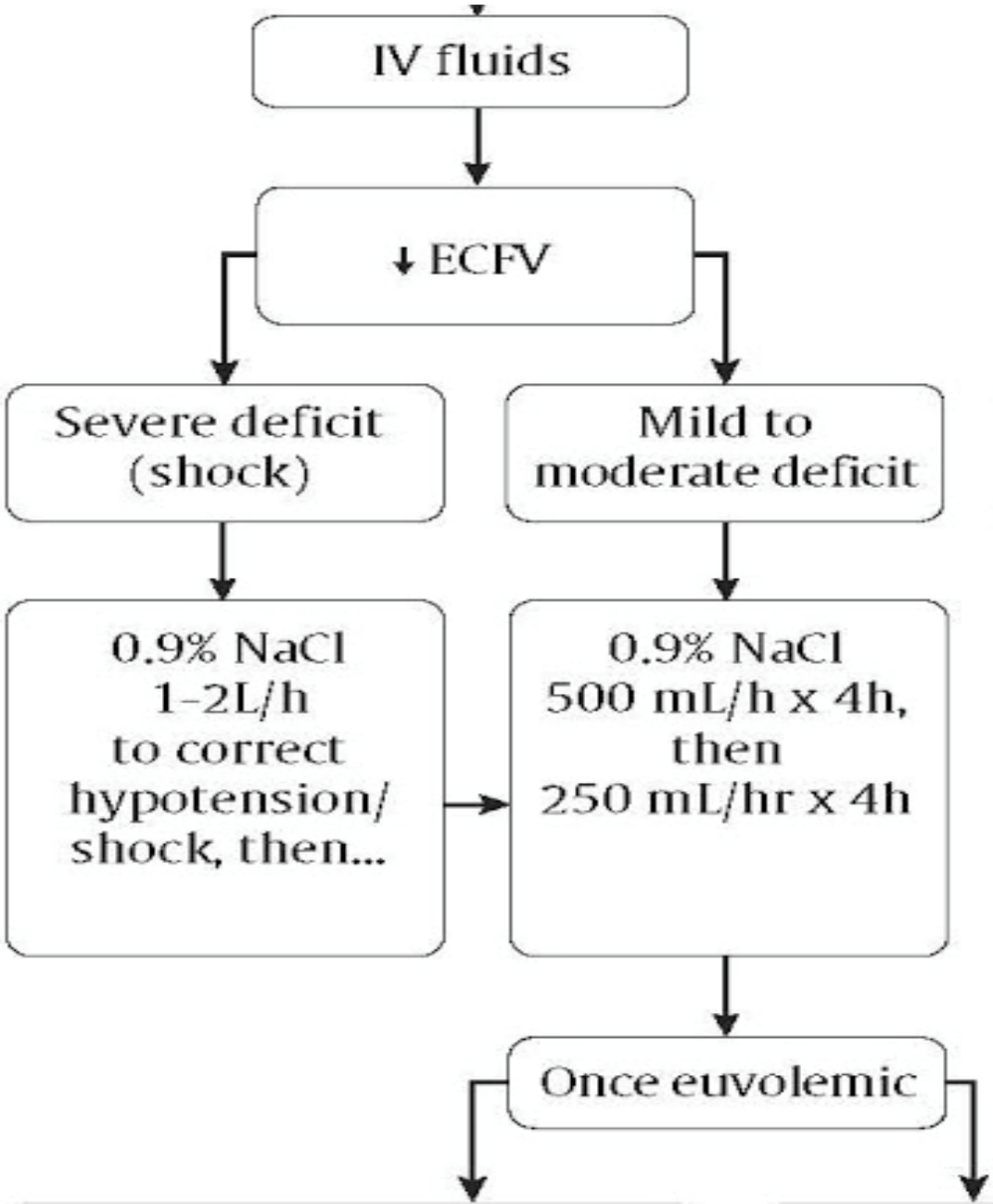
Replace Fluids with IV 0.9% NaCl until Euvolemic



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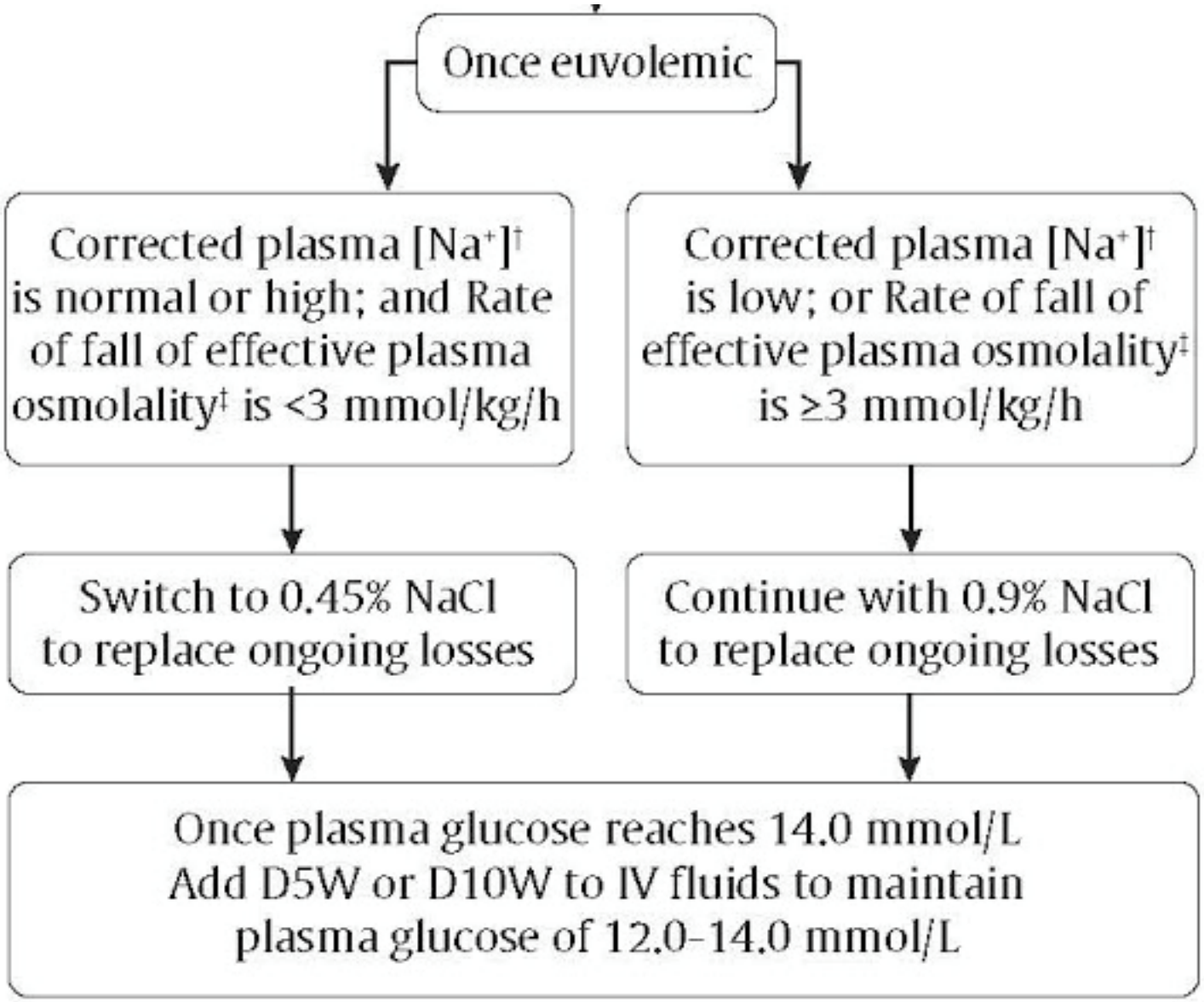
Once euvolemic, consider plasma Na^+ and glucose to determine IV fluid type



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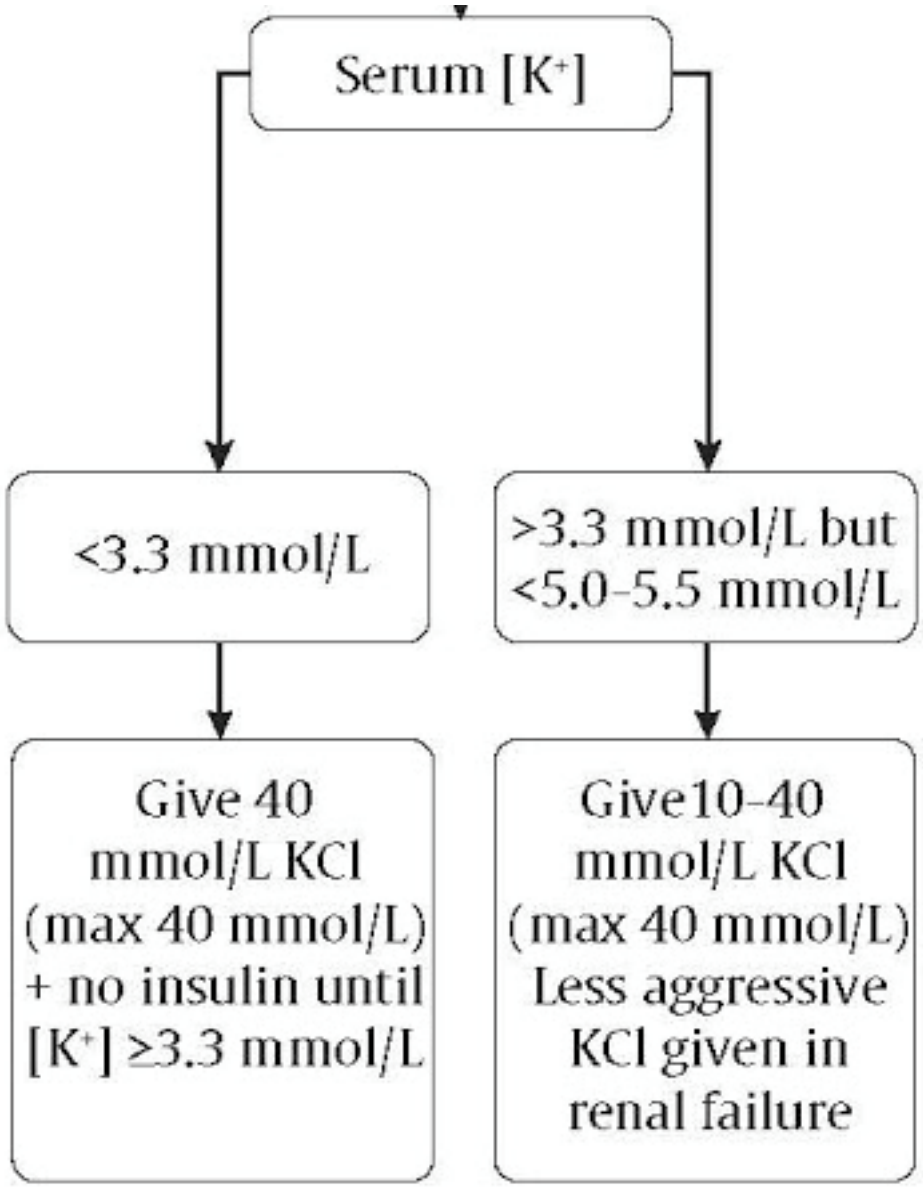
Replace Potassium: Hypokalemia is an avoidable cause of death in DKA



Correct K^+ first
THEN
start insulin

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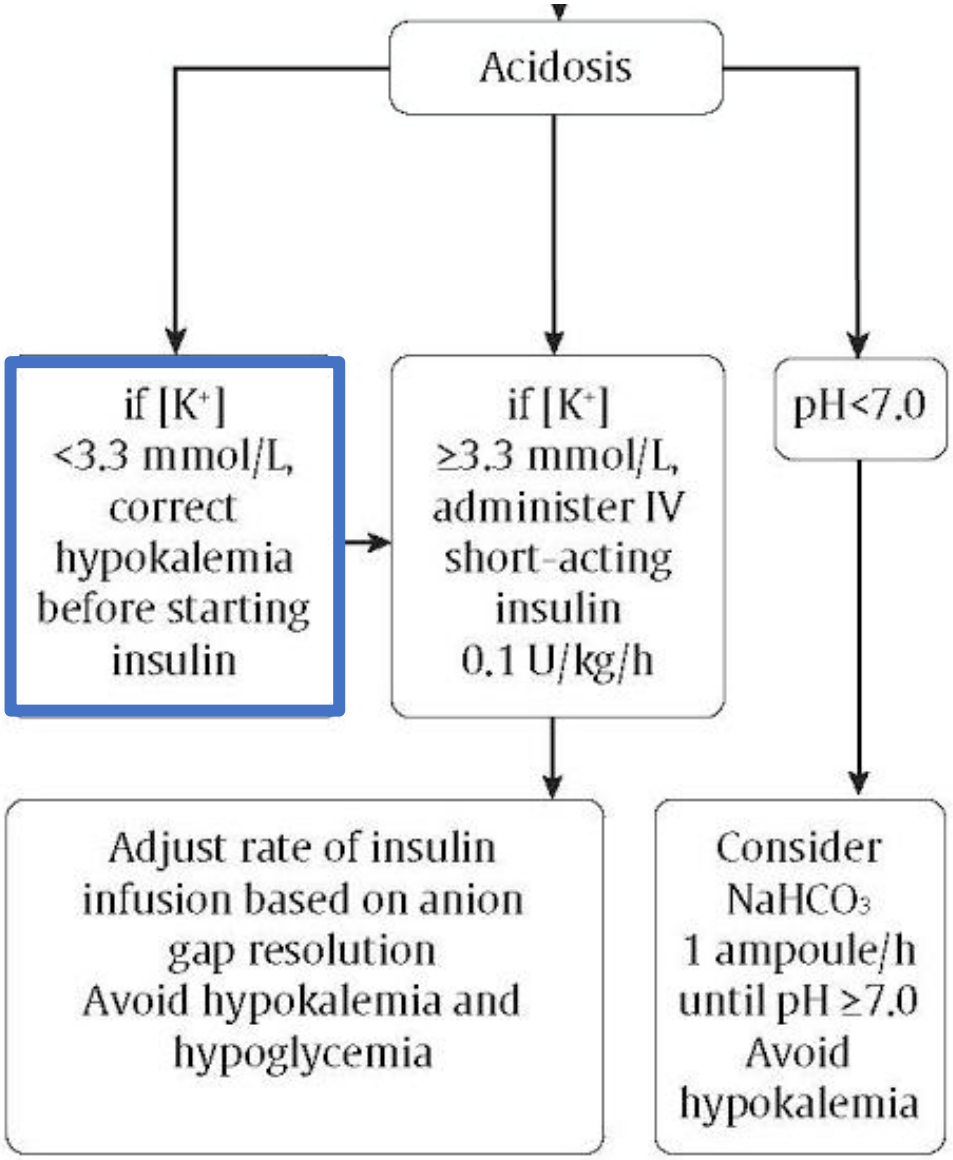
Management of Acidosis with Insulin



Insulin should be maintained until the anion gap normalizes

Insulin used to treat the acidosis, not the glucose!

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Identify and Treat the Precipitating Factor

- Insulin omission – **MOST COMMON CAUSE of DKA**
- New diagnosis of diabetes
- Infection / Sepsis
- Myocardial infarction
  - Small rise in troponin may occur without overt ischemia
  - ECG changes may reflect hyperkalemia
- Thyrotoxicosis
- Drugs

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DKA, diabetic ketoacidosis

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Prevention of DKA / HHS

- **Type 1 diabetes**
  - Education around sick day management
  - Continuation of insulin even when not eating
  - Frequent monitoring when ill
- **Type 2 diabetes**
  - Education around sick day management
  - Frequent monitoring when ill

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DKA, diabetic ketoacidosis;, *HHS*, hyperosmolar hyperglycemic state

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Priorities* to be addressed in the management of adults presenting with hyperglycemic emergencies		
Metabolic	**Precipitating cause of DKA/HHS**	**Other complications of DKA/HHS**
• ECFV contraction • Potassium deficit and abnormal concentration • Metabolic acidosis • Hyperosmolality (water deficit leading to increased corrected sodium concentration plus hyperglycemia)	• New diagnosis of diabetes • Insulin omission • Infection • Myocardial infarction • Stroke • ECG changes may reflect hyperkalemia • A small increase in troponin may occur without overt ischemia • Thyrotoxicosis • Trauma • Drugs	• Hyper/hypokalemia • ECFV overexpansion • Cerebral edema • Hypoglycemia • Pulmonary emboli • Aspiration • Hypocalcemia (if phosphate used) • Stroke • Acute renal failure • Deep vein thrombosis

*Severity of issue will dictate priority of action

DKA, diabetic ketoacidosis; ECFV, extracellular fluid volume; HHS, hyperosmolar hyperglycemic state

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Recommendation 1

1. In adults with DKA or HHS, a **protocol** should be followed that incorporates the following principles of treatment **fluid** resuscitation, avoidance of **hypokalemia**, **insulin** administration, avoidance of rapidly falling serum osmolality, and search for **precipitating** cause (as illustrated in Figure 1) [Grade D, Consensus]

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DKA, diabetic ketoacidosis; *HHS*, hyperosmolar hyperglycemic state

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Recommendation 2

2. **Point-of-care capillary beta-hydroxybutyrate** may be measured in the hospital or outpatient setting [Grade D, Level 4] in adults with type 1 diabetes with CBG >14.0 mmol/L to screen for DKA, and a **beta-hydroxybutyrate >1.5 mmol/L** warrants further testing for DKA [Grade B, Level 2]. Negative urine ketones should not be used to rule out DKA [Grade D, Level 4]

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CBG, capillary blood glucose*; DKA,* diabetic ketoacidosis

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Recommendation 3

3. In adults with DKA, intravenous **0.9% sodium chloride** should be administered initially at 500 mL/h for 4 hours, then 250 mL/h for 4 hours [Grade B, Level 2] with consideration of a higher initial rate (1-2 L/h) in the presence of shock [Grade D, Consensus]. For adults with HHS, intravenous fluid administration should be individualized [Grade D, Consensus]

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DKA, diabetic ketoacidosis; *HHS*, hyperosmolar hyperglycemic state

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Recommendation 4

4. In adults with DKA, an infusion of short-acting intravenous **insulin** of **0.10 units/kg/h** should be used [Grade B, Level 2]. The insulin infusion rate should be maintained until the **resolution of ketosis** [Grade B, Level 2] as measured by the normalization of the plasma anion gap [Grade D, Consensus]. Once the PG concentration **falls to 14.0 mmol/L, intravenous dextrose** should be started to avoid hypoglycemia [Grade D, Consensus]

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DKA, diabetic ketoacidosis; *PG*, plasma glucose

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Recommendation 5

5. Individuals treated with **SGLT2 inhibitors** with symptoms of **DKA** should be assessed for this condition **even if BG is not elevated** [Grade D, Consensus]

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DKA, diabetic ketoacidosis

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Key Messages

- Diabetic ketoacidosis (DKA) and hyperosmolar hyperglycemic state (HHS) should be **suspected in ill persons with diabetes**. If either DKA or HHS is diagnosed, **precipitating factors** must be sought and treated

- DKA and HHS are **medical emergencies** that require treatment and monitoring for multiple metabolic abnormalities and vigilance for complications

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Key Messages

- A **normal** or mildly elevated blood **glucose does not rule out** diabetic ketoacidosis in certain conditions such as **pregnancy** or with **SGLT2 inhibitor** use

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Key Messages

- DKA requires **intravenous insulin** administration (0.1 units/kg/h) for resolution; bicarbonate therapy may be considered only for extreme acidosis (pH ≤7.0)

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DKA, diabetic ketoacidosis

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Key Messages for People with Diabetes

When you are sick, your blood glucose levels may fluctuate and be unpredictable:

- During these times, it is a good idea to **check your blood glucose levels more often** than usual (for example, every 2 to 4 hours)
- **Drink** plenty of sugar-free fluids or water
- If you have **type 1 diabetes** with blood glucose levels remaining **over 14 mmol/L** before meals, or if you have **symptoms of diabetic ketoacidosis** (see chapter) **check for ketones** by performing a urine ketone test or blood ketone test. **Blood ketone testing** is preferred over urine testing

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Key Messages for People with Diabetes

- Develop a **"Sick Day" plan** with your diabetes healthcare team. This should include information on:
  - which diabetes medications you should continue and which ones you should temporarily stop;
  - guidelines for insulin adjustment if you are on insulin; and
  - advice on when to contact your health-care provider or go to the emergency room

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CLINICAL PRACTICE GUIDELINES
The Canadian Diabetes Association has become Diabetes Canada*

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Diabetes Canada is helping you provide patient-centred diabetes care and chronic disease management.

Coming soon! New Guidelines April 2018

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Frequently used healthcare provider tools and resources

- 2017 PHYSICAL ACTIVITY DECISION TOOL
- SCREENING FOR AND DIAGNOSING DIABETES
- SELF-MONITORING BLOOD GLUCOSE
- REDUCING VASCULAR RISK
- 2016 PHARMACOTHERAPY FOR TYPE 2 DIABETES
- INDIVIDUALIZING YOUR PATIENT'S A1C TARGET

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News & Highlights

The Canadian Diabetes Association has changed it's name to Diabetes Canada! Find out more

November 2016 Interim Update to the Guidelines
Updated recommendations to the Pharmacologic Management of Type 2 Diabetes

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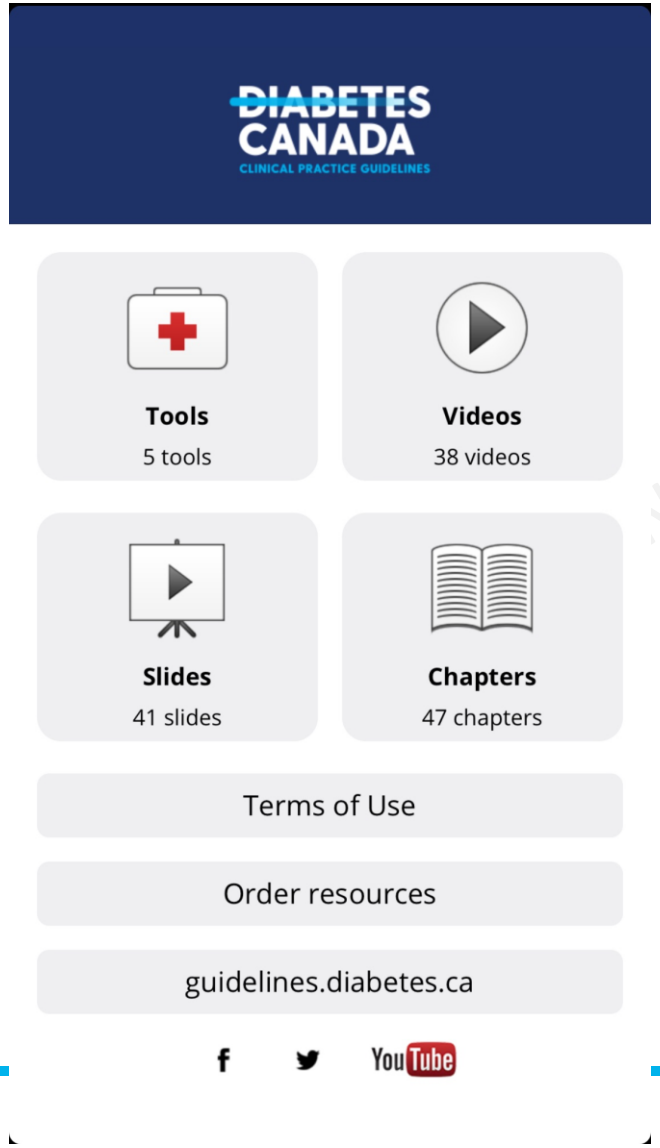

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Available on the App Store

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Diabetes Canada Clinical Practice Guidelines

http://guidelines.diabetes.ca – for health-care providers

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1-800-BANTING (226-8464)

http://diabetes.ca – for people with diabetes

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